

Birthday Dictionaries 🎂

Exercise 33 (and [Solution](#))

This exercise is Part 1 of 4 of the birthday data exercise series. The other exercises are: [Part 2](#), [Part 3](#), and [Part 4](#).

For this exercise, we will keep track of when our friend's birthdays are, and be able to find that information based on their name. Create a dictionary (in your file) of names and birthdays. When you run your program it should ask the user to enter a name, and return the birthday of that person back to them. The interaction should look something like this:

```
>>> Welcome to the birthday dictionary. We know the birthdays of:
Albert Einstein
Benjamin Franklin
Ada Lovelace
>>> Who's birthday do you want to look up?
Benjamin Franklin
>>> Benjamin Franklin's birthday is 01/17/1706.
```

Happy coding!

Birthday Json 🐍🐍

Exercise 34 (and [Solution](#))

This exercise is Part 2 of 4 of the birthday data exercise series. The other exercises are: [Part 1](#), [Part 3](#), and [Part 4](#).

In the [previous exercise](#) we created a dictionary of famous scientists' birthdays. In this exercise, modify your program from [Part 1](#) to load the birthday dictionary from a JSON file on disk, rather than having the dictionary defined in the program.

Bonus: Ask the user for another scientist's name and birthday to add to the dictionary, and update the JSON file you have on disk with the scientist's name. If you run the program multiple times and keep adding new names, your JSON file should keep getting bigger and bigger.

Birthday Months 🐣 🐣

Exercise 35 (and [Solution](#))

This exercise is Part 3 of 4 of the birthday data exercise series. The other exercises are: [Part 1](#), [Part 2](#), and [Part 4](#).

In the [previous exercise](#) we saved information about famous scientists' names and birthdays to disk. In this exercise, load that JSON file from disk, extract the months of all the birthdays, and count how many scientists have a birthday in each month.

Your program should output something like:

```
{  
    "May": 3,  
    "November": 2,  
    "December": 1  
}
```

Birthday Plots 🎂🎂🎂

Exercise 36 (and [Solution](#))

This exercise is Part 4 of 4 of the birthday data exercise series. The other exercises are: [Part 1](#), [Part 2](#), and [Part 3](#).

In the [previous exercise](#) we counted how many birthdays there are in each month in our dictionary of birthdays.

In this exercise, use the [bokeh](#) Python library to plot a histogram of which months the scientists have birthdays in! Because it would take a long time for you to input the months of various scientists, you can use [my scientist birthday JSON file](#). Just parse out the months (if you don't know how, I suggest looking at the [previous exercise](#) or [its solution](#)) and draw your histogram.

If you are using a purely web-based interface for coding, this exercise won't work for you, since it requires installing the bokeh Python package. Now might be a good time to [install Python on your own computer](#).